

Ryan G. Coe *Curriculum Vitae*

Associate Professor

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Education

- **Ph.D. in Aerospace Engineering (2013)**

Virginia Polytechnic Institute and State University (Virginia Tech), Blacksburg, VA

Advisor: Wayne L. Neu

Dissertation Title: “Improved Underwater Vehicle Control and Maneuvering Analysis with Computational Fluid Dynamics Simulations”

- **B.S. in Ocean Engineering (2009)**

Virginia Polytechnic Institute and State University (Virginia Tech), Blacksburg, VA

Graduated Magna Cum Laude

Publications

Journal articles

- J55. Alicia Keow, Jantzen Lee, Giorgio Bacelli, and **Ryan G. Coe**. Comparative analysis of pendulum and flywheel power take-off mechanisms for wave energy conversion. *IEEE Transactions on Energy Conversion*, pages 1–12, 2026. ISSN 1558-0059. doi: 10.1109/TEC.2026.3652091. URL <https://ieeexplore.ieee.org/document/11340697>
- J54. Dominic D. Forbush, **Ryan G. Coe**, Giorgio Bacelli, Daniel T. Gaebele, and Alicia Keow. Towards an intuitive application of WEC control co-design. *Ocean Engineering*, 341:122762, 2025. ISSN 0029-8018. doi: 10.1016/j.oceaneng.2025.122762. URL <https://www.sciencedirect.com/science/article/pii/S002980182502445X>
- J53. Daniel T. Gaebele, **Ryan G. Coe**, Giorgio Bacelli, Thomas Lanagan, Paul Fucile, Umesh A. Korde, and John Toole. Powering the Woods Hole X-Spar buoy with ocean wave energy—a control co-design feasibility study. *Energies*, 18(16), 2025. ISSN 1996-1073. doi: 10.3390/en18164442. URL <https://www.mdpi.com/1996-1073/18/16/4442>
- J52. Jian Tan, **Ryan G. Coe**, and George Lavidas. Benchmark of numerical modeling approaches on the systematic performance evaluation of wave energy converters. *Applied Ocean Research*, 162:104725, 2025. ISSN 0141-1187. doi: 10.1016/j.apor.2025.104725. URL <https://www.sciencedirect.com/science/article/pii/S0141118725003116>
- J51. **Ryan G. Coe**, Giorgio Bacelli, Daniel Gaebele, Alicia Keow, and Dominic Forbush. Co-design of a wave energy converter through bi-conjugate impedance matching. *Mechatronics*, 111:103395, 2025. ISSN 0957-4158. doi: 10.1016/j.mechatronics.2025.103395. URL <https://www.sciencedirect.com/science/article/pii/S0957415825001047>
- J50. Alicia Keow, Jantzen Lee, Giorgio Bacelli, and **Ryan G. Coe**. Design principles for resonant wave energy converters: Benchmarking power capture and flow. *IEEE Transactions on Energy Conversion*, pages 1–12, 2025. ISSN 1558-0059. doi: 10.1109/TEC.2025.3593152. URL <https://ieeexplore.ieee.org/document/11097361>
- J49. Nicholas Ross, Delaney Heileman, A. Gerrit Motes, Anwi Fomukong, Giorgio Bacelli, Steven J. Spencer, Dominic D. Forbush, Kevin Dullea, and **Ryan G. Coe**. A portable wave tank and wave energy converter for engineering dissemination and outreach. *Hardware*, 3(2), 2025. ISSN 2813-6640. doi: 10.3390/hardware3020005. URL <https://www.mdpi.com/2813-6640/3/2/5>

- J48. Tanvir Alam Shifat, **Ryan Coe**, Giorgio Bacelli, and Ted Brekken. Real-time sea state estimation for wave energy converter control via machine learning. *Applied Sciences*, 15(10), 2025. ISSN 2076-3417. doi: 10.3390/app15105772. URL <https://www.mdpi.com/2076-3417/15/10/5772>
- J47. Jeff Grasberger, Dominic Forbush, Johannes Spinneken, Mark Bruggemann, Jantzen Lee, Alex Franks, John Reine, Giorgio Bacelli, Albert Plueddemann, Donald Peters, and **Ryan G. Coe**. Hydrodynamic characterization of the Coastal Pioneer Array ocean observing system. *Journal of Ocean Engineering and Marine Energy*, 2025. doi: 10.1007/s40722-025-00392-y. URL <https://doi.org/10.1007/s40722-025-00392-y>
- J46. Ding Peng Liu, Lance Manuel, and **Ryan G. Coe**. Toward extending the life of a floating offshore wind turbine using sheltering from upstream wave energy converters. *Journal of Offshore Mechanics and Arctic Engineering*, pages 1–19, 04 2025. ISSN 0892-7219. doi: 10.1115/1.4068521. URL <https://doi.org/10.1115/1.4068521>
- J45. Jantzen Lee, Alicia Keow, **Ryan G. Coe**, Giorgio Bacelli, Johannes Spinneken, Steven J. Spencer, and Damian Gallegos-Patterson. Theory, analysis, and testing of an angular resonator for wave energy generation. *Journal of Ocean Engineering and Marine Energy*, 11(1):97–107, 2025. doi: 10.1007/s40722-024-00366-6. URL <https://link.springer.com/article/10.1007/s40722-024-00366-6>
- J44. Chris Dizon, **Ryan G. Coe**, Andrew Hamilton, Dominic D. Forbush, Michael Anderson, Ted Brekken, and Giorgio Bacelli. Analysis on Evaluations of Monterey Bay Aquarium Research Institute’s Wave Energy Converter’s Field Data Using WEC-Sim and Gazebo: A Simulation Tool Comparison. *Applied Sciences*, 14(23), 2024. ISSN 2076-3417. doi: 10.3390/app142311169. URL <https://www.mdpi.com/2076-3417/14/23/11169>
- J43. Michael C. Devin, Daniel T. Gaebele, Carlos A. Michelén Ströfer, Jeff T. Grasberger, Jantzen Lee, **Ryan G. Coe**, and Giorgio Bacelli. High-dimensional control co-design of a wave energy converter with a novel pitch resonator power takeoff system. *Ocean Engineering*, 312:119124, September 2024. ISSN 0029-8018. doi: 10.1016/j.oceaneng.2024.119124. URL <https://www.sciencedirect.com/science/article/pii/S0029801824024624>
- J42. Alfred Cotten, Adi Kurniawan, Vincent S. Neary, **Ryan G. Coe**, and Giorgio Bacelli. A compressible degree of freedom as a means for improving the performance of heaving wave energy converters. *Renewable Energy*, page 120421, 2024. ISSN 0960-1481. doi: 10.1016/j.renene.2024.120421. URL <https://www.sciencedirect.com/science/article/pii/S0960148124004865>
- J41. Simone Giorgi, **Ryan G. Coe**, Meagan Reasoner, Giorgio Bacelli, Dominic Forbush, Scott Jensen, and Andrew Hamilton. Wave energy converter power take-off modeling and validation from experimental bench tests. *IEEE Journal of Oceanic Engineering*, 49(2):446–457, 2024. doi: 10.1109/JOE.2023.3345903. URL <https://ieeexplore.ieee.org/document/10436658>
- J40. Umesh A. Korde, L. Andrew Gish, Giorgio Bacelli, and **Ryan G. Coe**. Wave energy conversion using a small tubular free-floating device. *Journal of Ocean Engineering and Marine Energy*, 2023. doi: 10.1007/s40722-023-00299-6. URL <https://doi.org/10.1007/s40722-023-00299-6>
- J39. John V. Ringwood, Nathan Tom, Francesco Ferri, Yi-Hsiang Yu, **Ryan G. Coe**, Kelley Ruehl, Giorgio Bacelli, Shuo Shi, Ron J. Patton, Paolino Tona, Guillaume Sabiron, Alexis Merigaud, Bradley A. Ling, and Nicolas Faedo. The wave energy converter control competition (WECCOMP): Wave energy control algorithms compared in both simulation and tank testing. *Applied Ocean Research*, 138:103653, September 2023. ISSN 0141-1187. doi: 10.1016/j.apor.2023.103653. URL <https://www.sciencedirect.com/science/article/pii/S0141118723001943>
- J38. Carlos A. Michelén Ströfer, Daniel T. Gaebele, **Ryan G. Coe**, and Giorgio Bacelli. Control co-design of power take-off systems for wave energy converters using WecOptTool. *IEEE Transactions on Sustainable Energy*, 14(4):2157–2167, 2023. doi: 10.1109/TSTE.2023.3272868. URL <https://ieeexplore.ieee.org/document/10114969>
- J37. Jorge A. Leon-Quiroga, Giorgio Bacelli, Dominic D. Forbush, Steven J. Spencer, and **Ryan G. Coe**. An efficient and effective WEC power take-off system. *IEEE Transactions on Sustainable Energy*, 14(3):1526–1539, July 2023. doi: 10.1109/TSTE.2023.3237125. URL <https://ieeexplore.ieee.org/document/10017387>
- J36. Dominic D. Forbush, Giorgio Bacelli, Steven J. Spencer, **Ryan G. Coe**, David G. Wilson, and Bryson Robertson. A self-tuning WEC controller for changing sea states. *International Marine Energy Journal*, 5(3): 327–338, Dec. 2022. doi: 10.36688/imej.5.327-338. URL <https://marineenergyjournal.org/imej/article/view/70>

- J35. **Ryan G. Coe**, Lance Manuel, and Andreas F. Haselsteiner. On limiting the influence of serial correlation in metocean data for prediction of extreme return levels and environmental contours. *Ocean Engineering*, 266:113032, December 2022. ISSN 0029-8018. doi: 10.1016/j.oceaneng.2022.113032. URL <https://www.sciencedirect.com/science/article/pii/S0029801822023150>
- J34. Bret Bosma, **Ryan Coe**, Giorgio Bacelli, Ted Brekken, and Budi Gunawan. Mini-DAQ: A lightweight, low-cost, high resolution, data acquisition system for wave energy converter testing. *HardwareX*, 12:e00332, October 2022. ISSN 2468-0672. doi: 10.1016/j.ohx.2022.e00332. URL <https://www.sciencedirect.com/science/article/pii/S2468067222000773>
- J33. Dominic D. Forbush, Giorgio Bacelli, and **Ryan G. Coe**. Self-tuning, load-mitigating feedback control of a 3-DOF point absorber. *International Marine Energy Journal*, 5(1):23–35, June 2022. doi: 10.36688/imej.5.23-35. URL <https://marineenergyjournal.org/imej/article/view/68>
- J32. Ali S. Haider, Ted K. A. Brekken, **Ryan G. Coe**, Giorgio Bacelli, and Alan McCall. On real-time hybrid testing of ocean wave energy conversion systems: An experimental study. *IEEE Open Journal of Industry Applications*, 3:30–40, February 2022. doi: 10.1109/OJIA.2022.3148388. URL <https://ieeexplore.ieee.org/document/9705552/>
- J31. Liliana Haus, D. Todd Griffith, **Ryan G. Coe**, and Giorgio Bacelli. Development and characterization of a coupled structural dynamics model for the Sandia wave energy converter testbed. *Journal of Ocean Engineering and Marine Energy*, 8:117–135, January 2022. doi: 10.1007/s40722-021-00220-z. URL <https://doi.org/10.1007/s40722-021-00220-z>
- J30. Dominic D. Forbush, Giorgio Bacelli, Steven J. Spencer, **Ryan G. Coe**, Bret Bosma, and Pedro Lomonaco. Design and testing of a free floating dual flap wave energy converter. *Energy*, 240:122485, February 2022. ISSN 0360-5442. doi: 10.1016/j.energy.2021.122485. URL <https://www.sciencedirect.com/science/article/pii/S0360544221027341>
- J29. Nataliia Y. Sergiienko, Giorgio Bacelli, **Ryan G. Coe**, and Benjamin S. Cazzolato. A comparison of efficiency-aware model-predictive control approaches for wave energy devices. *Journal of Ocean Engineering and Marine Energy*, 8:17–29, February 2022. doi: 10.1007/s40722-021-00214-x. URL <https://doi.org/10.1007/s40722-021-00214-x>
- J28. Andreas F. Haselsteiner, **Ryan G. Coe**, Lance Manuel, Wei Chai, Bernt Leira, Guilherme Clarindo, C. Guedes Soares, Ásta Hannesdóttir, Nikolay Dimitrov, Aljoscha Sander, Jan-Hendrik Ohlendorf, Klaus-Dieter Thoben, Guillaume de Hauteclocque, Ed Mackay, Philip Jonathan, Chi Qiao, Andrew Myers, Anna Rode, Arndt Hildebrandt, Boso Schmidt, Erik Vanem, and Arne Bang Huseby. A benchmarking exercise for environmental contours. *Ocean Engineering*, 236:109504, 2021. ISSN 0029-8018. doi: 10.1016/j.oceaneng.2021.109504. URL <https://www.sciencedirect.com/science/article/pii/S0029801821009033>
- J27. Andrew Hamilton, François Cazenave, Dominic Forbush, **Ryan G. Coe**, and Giorgio Bacelli. The MBARI-WEC: a power source for ocean sensing. *Journal of Ocean Engineering and Marine Energy*, 7(2):189–200, 2021. doi: 10.1007/s40722-021-00197-9. URL <https://doi.org/10.1007/s40722-021-00197-9>
- J26. **Ryan G. Coe**, Seongho Ahn, Vincent S. Neary, Peter H. Kobos, and Giorgio Bacelli. Maybe less is more: Considering capacity factor, saturation, variability, and filtering effects of wave energy devices. *Applied Energy*, 291:116763, 2021. ISSN 0306-2619. doi: 10.1016/j.apenergy.2021.116763. URL <https://www.sciencedirect.com/science/article/pii/S0306261921002701>
- J25. **Ryan G. Coe**, Giorgio Bacelli, and Dominic Forbush. A practical approach to wave energy modeling and control. *Renewable and Sustainable Energy Reviews*, 142:110791, 2021. ISSN 1364-0321. doi: 10.1016/j.rser.2021.110791. URL <https://www.sciencedirect.com/science/article/pii/S1364032121000861>
- J24. Pablo Roperro-Giralda, Alejandro J. C. Crespo, **Ryan G. Coe**, Bonaventura Tagliaferro, Jose M. Dominguez, Giorgio Bacelli, and Moncho Gomez-Gesteira. Modelling a heaving point-absorber with a closed-loop control system using the DualSPHysics code. *Energies*, 14(3), 2021. ISSN 1996-1073. doi: 10.3390/en14030760. URL <https://www.mdpi.com/1996-1073/14/3/760>

- J23. Aubrey Eckert, Nevin Martin, **Ryan G. Coe**, Bibiana Seng, Zacharia Stuart, and Zachary Morrell. Development of a comparison framework for evaluating environmental contours of extreme sea states. *Journal of Marine Science and Engineering*, 9(1):16, 2021. doi: 10.3390/jmse9010016. URL <https://doi.org/10.3390/jmse9010016>
- J22. Giorgio Bacelli and **Ryan G. Coe**. Comments on control of wave energy converters. *IEEE Transactions on Control Systems Technology*, 29(1):478–481, Jan 2021. doi: 10.1109/TCST.2020.2965916. URL <https://ieeexplore.ieee.org/document/9005201>
- J21. **Ryan G. Coe**, Giorgio Bacelli, Sterling Olson, Vincent S. Neary, and Matthew B. R. Topper. Initial conceptual demonstration of control co-design for WEC optimization. *Journal of Ocean Engineering and Marine Energy*, 6(4):441–449, 2020. doi: 10.1007/s40722-020-00181-9. URL <https://doi.org/10.1007/s40722-020-00181-9>
- J20. Umesh A. Korde, **Ryan G. Coe**, and Giorgio Bacelli. Deterministic incident wave elevation prediction in intermediate water depth. *Journal of Ocean Engineering and Marine Energy*, 6(4):359–376, November 2020. doi: 10.1007/s40722-020-00177-5. URL <https://doi.org/10.1007/s40722-020-00177-5>
- J19. Blake P. Driscoll, L. Andrew Gish, and **Ryan G. Coe**. A scoping study to determine the location-specific WEC threshold size for wave-powered AUV recharging. *IEEE Journal of Oceanic Engineering*, pages 1–10, 2020. ISSN 1558-1691. doi: 10.1109/JOE.2020.2973032. URL <https://ieeexplore.ieee.org/document/9044418>
- J18. David G. Wilson, Rush D. Robinett, Giorgio Bacelli, Ossama Abdelkhalik, and **Ryan G. Coe**. Extending complex conjugate control to nonlinear wave energy converters. *Journal of Marine Science and Engineering*, 8(2):84, Jan 2020. ISSN 2077-1312. doi: 10.3390/jmse8020084. URL <http://dx.doi.org/10.3390/jmse8020084>
- J17. Giorgio Bacelli, Victor Nevarez, **Ryan G. Coe**, and David Wilson. Feedback resonating control for a wave energy converter. *IEEE Industrial Automation and Control*, 56(2):1862–1868, 2020. doi: 10.1109/TIA.2019.2958018. URL <https://ieeexplore.ieee.org/document/8926523>
- J16. Hancheol Cho, Giorgio Bacelli, and **Ryan G. Coe**. Model predictive control tuning by inverse matching for a wave energy converter. *Energies*, 12(21):4158, Oct 2019. ISSN 1996-1073. doi: 10.3390/en12214158. URL <http://dx.doi.org/10.3390/en12214158>
- J15. Jennifer van Rij, Yi-Hsiang Yu, Yi Guo, and **Ryan G. Coe**. A wave energy converter design load case study. *Journal of Marine Science and Engineering*, 7(8):250, Jul 2019. ISSN 2077-1312. doi: 10.3390/jmse7080250. URL <http://dx.doi.org/10.3390/jmse7080250>
- J14. Phong T. T. Nguyen, Lance Manuel, and **Ryan G. Coe**. On the Development of an Efficient Surrogate Model for Predicting Long-Term Extreme Loads on a Wave Energy Converter. *Journal of Offshore Mechanics and Arctic Engineering*, 141(6), 03 2019. ISSN 0892-7219. doi: 10.1115/1.4042944. URL <https://doi.org/10.1115/1.4042944>. 061103
- J13. **Ryan G. Coe**, Brian J. Rosenberg, Eliot W. Quon, Chris C. Chartrand, Yi-Hsiang Yu, Jennifer van Rij, and Tim R. Mundon. CFD design-load analysis of a two-body wave energy converter. *Journal of Ocean Engineering and Marine Energy*, 5(2):99–117, 2019. doi: 10.1007/s40722-019-00129-8. URL <https://doi.org/10.1007/s40722-019-00129-8>
- J12. Giorgio Bacelli, Steven J. Spencer, David C. Patterson, and **Ryan G. Coe**. Wave tank and bench-top control testing of a wave energy converter. *Applied Ocean Research*, 86:351 – 366, 2019. ISSN 0141-1187. doi: 10.1016/j.apor.2018.09.009. URL <http://www.sciencedirect.com/science/article/pii/S0141118717307484>
- J11. Vincent S. Neary, **Ryan G. Coe**, João Cruz, Kevin Haas, Giorgio Bacelli, Yannick Debruyne, Seongho Ahn, and Victor Nevarez. Classification systems for wave energy resources and WEC technologies. *International Marine Energy Journal*, 1(2):71–79, Nov 2018. doi: 10.36688/imej.1.71-79. URL <https://marineenergyjournal.org/imej/article/view/29>
- J10. Samuel J. Edwards and **Ryan G. Coe**. The Effect of Environmental Contour Selection on Expected Wave Energy Converter Response. *Journal of Offshore Mechanics and Arctic Engineering*, 141(1), 08 2018. ISSN 0892-7219. doi: 10.1115/1.4040834. URL <https://doi.org/10.1115/1.4040834>. 011901

- J9. **Ryan G. Coe**, Carlos Michelen, Aubrey Eckert-Gallup, and Cédric Sallaberry. Full long-term design response analysis of a wave energy converter. *Renewable Energy*, 116:356 – 366, 2018. ISSN 0960-1481. doi: 10.1016/j.renene.2017.09.056. URL <http://www.sciencedirect.com/science/article/pii/S0960148117309187>
- J8. Lance Manuel, Phong T. T. Nguyen, Jarred Canning, **Ryan G. Coe**, Aubrey C. Eckert-Gallup, and Nevin Martin. Alternative approaches to develop environmental contours from metocean data. *Journal of Ocean Engineering and Marine Energy*, 4(4):293–310, 2018. doi: 10.1007/s40722-018-0123-0. URL <https://doi.org/10.1007/s40722-018-0123-0>
- J7. Ossama Abdelkhalik, Shangyan Zou, Rush Robinett, Giorgio Bacelli, David Wilson, and **Ryan Coe**. Control of three degrees-of-freedom wave energy converters using pseudo-spectral methods. *Journal of Dynamic Systems, Measurement, and Control*, 140(7):074501–074501–9, 01 2018. URL <http://dx.doi.org/10.1115/1.4038860>
- J6. **Ryan G. Coe**, Yi-Hsiang Yu, and Jennifer van Rij. A survey of WEC reliability, survival and design practices. *Energies*, 11(1):4, Dec 2017. ISSN 1996-1073. doi: 10.3390/en11010004. URL <http://dx.doi.org/10.3390/en11010004>
- J5. Shangyan Zou, Ossama Abdelkhalik, Rush Robinett, Umesh Korde, Giorgio Bacelli, David Wilson, and **Ryan Coe**. Model predictive control of parametric excited pitch-surge modes in wave energy converters. *International Journal of Marine Energy*, 19:32 – 46, 2017. ISSN 2214-1669. doi: 10.1016/j.ijome.2017.05.002. URL <http://www.sciencedirect.com/science/article/pii/S2214166917300504>
- J4. O. Abdelkhalik, S. Zou, R. Robinett III, G. Bacelli, D. Wilson, **R. Coe**, and U. A. Korde. Multi resonant feedback control of three-degree-of-freedom wave energy converters. *IEEE Transactions on Sustainable Energy*, PP(99):1–1, 2017. ISSN 1949-3029. doi: 10.1109/TSTE.2017.2692647. URL <http://ieeexplore.ieee.org/document/7895127/>
- J3. Giorgio Bacelli, **Ryan G. Coe**, David Patterson, and David Wilson. System identification of a heaving point absorber: Design of experiment and device modeling. *Energies*, 10(10):472, 2017. ISSN 1996-1073. doi: 10.3390/en10040472. URL <http://www.mdpi.com/1996-1073/10/4/472>
- J2. **Ryan G. Coe**, Giorgio Bacelli, David G. Wilson, Ossama Abdelkhalik, Umesh A. Korde, and Rush D. Robinett III. A comparison of control strategies for wave energy converters. *International Journal of Marine Energy*, 20(Supplement C):45 – 63, 2017. ISSN 2214-1669. doi: 10.1016/j.ijome.2017.11.001. URL <http://www.sciencedirect.com/science/article/pii/S2214166917300905>
- J1. Ossama Abdelkhalik, Rush Robinett, Shangyan Zou, Giorgio Bacelli, **Ryan Coe**, Diana Bull, David Wilson, and Umesh Korde. On the control design of wave energy converters with wave prediction. *Journal of Ocean Engineering and Marine Energy*, 2(4):473–483, 2016. ISSN 2198-6452. doi: 10.1007/s40722-016-0048-4. URL <http://dx.doi.org/10.1007/s40722-016-0048-4>

Book contributions

- B1. Krish Thiagarajan Sharman and Roger Basu. *Ocean Structures Design: From Oil & Gas to Blue Economy*. American Institute of Physics, 2021 (in prep) (case-study contributed by **Ryan G. Coe**)

Editorials

- E2. **Ryan G. Coe** and Giorgio Bacelli. Useful power maximization for wave energy converters. *Energies*, 16(1), 2023. ISSN 1996-1073. doi: 10.3390/en16010529. URL <https://www.mdpi.com/1996-1073/16/1/529>
- E1. Andreas F. Haselsteiner, **Ryan G. Coe**, and Lance Manuel. A second benchmarking exercise on estimating extreme environmental conditions. *Ocean Engineering*, page 109111, 2021. ISSN 0029-8018. doi: 10.1016/j.oceaneng.2021.109111. URL <https://www.sciencedirect.com/science/article/pii/S0029801821005473>

Peer reviewed conference papers

- C68. Bret Bosma, Courtney Beringer, **Ryan Coe**, Giorgio Bacelli, Daniel Gaebele, Dominic Forbush, and Bryson Robertson. System identification techniques applied to a laboratory scale wec point absorber. In *Proceedings of the 16th European Wave and Tidal Energy Conference (EWTEC2025)*, volume 16, Funchal, Portugal, September 2025. doi: 10.36688/ewtec-2025-826. URL <https://doi.org/10.36688/ewtec-2025-826>
- C67. Elaine Liu, Jantzen Lee, and **Ryan G. Coe**. Development and validation of an inverted pendulum wave energy converter model. In *UMERC Conference*, Corvallis, OR, August 2025. University Marine Energy Research Community (UMERC)
- C66. Tanvir Alam Shifat, **Ryan Coe**, Giorgio Bacelli, and Ted Brekken. Deep learning-driven control optimization for wave energy converters via real-time sea state prediction. In *Proceedings of UMERC+OREC 2025*, Corvallis, OR, August 2025
- C65. Daniel Gaebele, Giorgio Bacelli, **Ryan Coe**, Dominic Forbush, Courtney Beringer, Bret Bosma, Bryson Robertson, and Pedro Lomonaco. Streamlined experimental approach to system identification of WEC dynamics and excitation characteristics. In *International Conference on Ocean, Offshore and Arctic Engineering (OMAE2025)*, Vancouver, British Columbia, Canada, June 2025. ASME
- C64. Jeff T. Grasberger, **Ryan G. Coe**, Daniel T. Gaebele, Michael C. Devin, Carlos A. Michelén Ströfer, and Giorgio Bacelli. Frequency array and wave phase realizations for wave energy converter control optimization. In *International Conference on Ocean Energy (ICOE 2024)*, Melbourne, Australia, September 2024
- C63. **Ryan G. Coe**, Michael C. Devin, Carlos A. Michelén Ströfer, Jantzen Lee, Giorgio Bacelli, Alicia Keow, Daniel T. Gaebele, Jeff Grasberger, Steven J. Spencer, Johannes Spinneken, Vincent S. Neary, and Brek Meuris. Co-design of a wave energy converter for autonomous power. In *15th IFAC Conference on Control Applications in Marine Systems, Robotics and Vehicles (IFAC-CAMS)*, volume 58, pages 446–451, Blacksburg, VA, September 2024. IFAC. doi: 10.1016/j.ifacol.2024.10.094. URL <https://www.sciencedirect.com/science/article/pii/S2405896324018494>
- C62. Tanvir Alam Shifat, **Ryan Coe**, Giorgio Bacelli, and Ted K.A. Brekken. Constrained pseudo-PI linear control of a wave energy converter via model predictive control. In *15th IFAC Conference on Control Applications in Marine Systems, Robotics and Vehicles (IFAC-CAMS)*, volume 58, pages 398–403, Blacksburg, VA, September 2024. IFAC. doi: 10.1016/j.ifacol.2024.10.086. URL <https://www.sciencedirect.com/science/article/pii/S240589632401841X>
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- T10. **Ryan G. Coe** and Giorgio Bacelli. A summary of the Advanced WEC Dynamics and Control project. Technical Report SAND2019-14355, Sandia National Laboratories, Albuquerque, NM, Nov 2019. URL <https://www.osti.gov/biblio/1592944>
- T9. **Ryan Geoffrey Coe**, Alice Gillespie, Chris Clayton Chartrand, Mike Morrow, Mike Delos-Reyes, Fabian Wendt, Yi-Hsiang Yu, Tuba Oskan-Haller, Pedro Lomanaco, Jesse D. Roberts, Sterling Olson, and Craig Jones. An approach for shear-stress based scour prediction. Technical Report SAND2019-0865, Sandia National Laboratories, Albuquerque, NM, Jan 2019. URL <https://www.osti.gov/biblio/1493823>
- T8. **Ryan G. Coe**, Zachary Morrell, Giorgio Bacelli, Dominic Forbush, Hannah Mankle, Bryony DuPont, and Vincent S. Neary. WEC optimization tool scoping report. Technical Report SAND2019-7580, Sandia National Laboratories, Albuquerque, NM, June 2019
- T7. **Ryan G. Coe**, Giorgio Bacelli, Steven J. Spencer, and Hancheol Cho. Initial results from wave tank test of closed-loop WEC control. Technical Report SAND2018-12858, Sandia National Laboratories, Albuquerque, NM, Nov 2018. URL <https://doi.org/10.2172/1531328>
- T6. **Ryan G. Coe**, Giorgio Bacelli, Victor Nevarez, Hancheol Cho, and Felipe Wilches-Bernal. A comparative study on wave prediction for WECs. Technical Report SAND2018-10945, Sandia National Laboratories, Albuquerque, NM, 2018. URL <https://www.osti.gov/biblio/1476166>
- T5. Jarred Canning, Samuel Edwards, Tyler Richard Esterly, Bibiana Elisabeth Seng, Laura Smith, Zacharia William Stuart, Aubrey Celia Eckert, Nevin Martin, and **Ryan G. Coe**. WEC extreme conditions modeling sandia summer intern 2017 report. Technical Report SAND2017-10592, Sandia National Laboratories, Albuquerque, NM, 2017. URL <https://www.osti.gov/biblio/1596200>
- T4. Giorgio Bacelli and **Ryan G. Coe**. State estimation for wave energy converters. Technical Report SAND2017-4401, Sandia National Laboratories, Albuquerque, NM, April 2017. URL <https://www.osti.gov/biblio/1365524>

- T3. **Ryan G. Coe**, Giorgio Bacelli, David Patterson, and David G. Wilson. Advanced WEC Dynamics & Controls FY16 testing report. Technical Report SAND2016-10094, Sandia National Labs, Albuquerque, NM, October 2016. URL <https://doi.org/10.2172/1330189>
- T2. David Wilson, Giorgio Bacelli, **Ryan G. Coe**, Diana L. Bull, Ossama Abdelkhalik, Umesh A. Korde, and Rush D. Robinett III. A comparison of WEC control strategies. Technical Report SAND2016-4293, Sandia National Labs, Albuquerque, NM, April 2016. URL <https://www.osti.gov/biblio/1431291>
- T1. **Ryan G. Coe**, Vincent S. Neary, Michael J. Lawson, Yi-Hsiang Yu, and Jochem Weber. Extreme conditions modeling workshop report. Technical Report SAND2014-16384R, Sandia National Laboratories, Albuquerque, NM, Jul 2014. URL <https://www.osti.gov/biblio/1149224>

Patents

Granted

- P6. Umesh Korde, **Ryan G. Coe**, and Giorgio Bacelli. Tubular wave power device. URL <https://patents.google.com/patent/US12129820B1>
- P5. Giorgio Bacelli, Dominic Forbush, Steven J. Spencer, **Ryan G. Coe**, and David G. Wilson. A self-tuning WEC controller for changing sea states. URL <https://patents.google.com/patent/US11703027B1>
- P4. David G. Wilson, Giorgio Bacelli, **Ryan Geoffrey Coe**, Rush D. Robinett, and Ossama Abdelkhalik. Nonlinear hydrostatic control of a wave energy converter. URL <https://patents.google.com/patent/US11326574B2>
- P3. Ossama Abdelkhalik, Rush D. Robinett, Shangyan Zou, David G. Wilson, Giorgio Bacelli, Umesh Korde, and **Ryan G. Coe**. Model predictive control of parametric excited pitch-surge modes in wave energy converters. URL <https://patents.google.com/patent/US10415537B2>
- P2. Ossama Abdelkhalik, Giorgio Bacelli, Shangyan Zou, Rush D. Robinett, David G. Wilson, and **Ryan G. Coe**. Pseudo-spectral method to control three-degree-of-freedom wave energy converters. URL <https://patents.google.com/patent/US10344736B2>
- P1. Ossama Abdelkhalik, Shangyan Zou, Rush D. Robinett, David G. Wilson, Giorgio Bacelli, **Ryan Geoffrey Coe**, and Umesh Korde. Multi-resonant feedback control of multiple degree-of-freedom wave energy converters. URL <https://patents.google.com/patent/US10488828B2>

Pending

- Giorgio Bacelli, Jantzen Lee, Steve Spencer, **Ryan Coe**, Alicia Keow, Kevin Dullea, Johannes Spinneken, Damian Gallegos-Patterson, Tim Donnelly, Al Plueddemann, John Reine, Derek Buffitt, Alex Franks, and Umesh Korde. Pitch resonator wave energy converter

Service, honors, & certificates

- **Editor/organizer:**
 - Associate Editor (2017-present), *Journal of Waterway, Port, Coastal, and Ocean Engineering*, <https://ascelibrary.org/journal/jwped5>
 - Section Board Member (2021-present), *Energies*, <https://www.mdpi.com/journal/energies>
 - Associate Editor (2024-present), *Journal of Ocean Engineering and Marine Energy*, <https://link.springer.com/journal/40722>

- Associate Editor (2024-present), *Journal of Offshore Mechanics and Arctic Engineering*, <https://asmedigitalcollection.asme.org/offshoremechanics>
- Convener (2020-2024), IEC TC 114/MT 62600-2 (“Design requirements for marine energy systems”), https://www.iec.ch/dyn/www/f?p=103:14:5246482804475:::FSP_ORG_ID,FSP_LANG_ID:5893,25
- Session organizer (2022), Design and Control of Ocean Wave Energy Converters, The 16th International Conference on Motion and Vibration Control (MoViC 2022), <https://ifacms-movic2022.seas.ucla.edu/home/>
- Guest Editor (2020-2022), Special Issue on Probabilistic Long-Term Extreme Conditions using Metocean Data, *Ocean Engineering*, <https://www.journals.elsevier.com/ocean-engineering/call-for-papers/probabilistic-long-term-extreme-conditions>
- Co-organizer (2018-present), A Benchmarking Exercise on Estimating Extreme Environmental Conditions (“EC Benchmark”), <https://github.com/ec-benchmark-organizers/ec-benchmark>
- Co-organizer (2017-2020), WEC Control Competition (WECCOMP), <http://www.eeng.nuim.ie/coer/wec-control-competition-released/>
- Topic Organizer (2017, 2025), Session Chair (2015, 2016, 2017, 2018, 2019), ASME Offshore Mechanics and Arctic Engineering Conference (OMAE)
- **Societies/committees:**
 - Subject-Matter Expert (2016-2020), IEC TC 114/MT 62600-2 (“Design requirements for marine energy systems”)
 - Member (2006-2019), SNAME/ASNE
 - T&R panel member (2014-2019), SNAME EC-12 (“Ocean Renewable Energy”)
- **Paper reviewer:** *Applied Ocean Research; Applied Energy; Applied Sciences; Electronics; Fluids; International Journal of Marine Energy; Energies; Ocean Engineering; Journal of Offshore Mechanics and Arctic Engineering; IEEE Transactions on Sustainable Energy; IEEE Transactions on Control Systems Technology; IEEE Transactions on Robotics; IEEE/ASME Transactions on Mechatronics; Journal of Fluids and Structures; Journal of Ocean Engineering and Marine Energy; Journal of Marine Science and Engineering; Renewable and Sustainable Energy Reviews; Marine Energy Technology Symposium (METS); Offshore Mechanics and Arctic Engineering Conference (OMAE); European Wave and Tidal Energy Conference (EWTEC); IFAC Conference on Control Applications in Marine Systems, Robotics, and Vehicles (CAMS)*
- **Honors:** IEC 1906 award (2024); Sandia Up & Coming Innovators Award (2017); Pratt Graduate Fellowship (2011, 2012); Virginia Space Grant Consortium Graduate Fellowship (2011, 2012); American Society of Naval Engineers (ASNE) Graduate Scholarship (2009); Society of Naval Architects and Marine Engineers (SNAME) Undergraduate Scholarship (2008); Virginia Tech College of Engineering Scholarship (2006)
- **Certificates:**
 - Virginia Tech Future Professoriate Certificate (2013)
 - Engineer in Training (2009)

Invited seminars, teaching

- Lead instructor, Introduction to Ocean Engineering (AOE2204), Virginia Tech, Blacksburg, VA, 2012.
- Workshops, seminars, & guest lectures
 - “Ocean Renewable Energy,” with Cesar Castillo, remote lecture for Duke University, September 10, 2025.
 - “Control and system design of wave energy converters,” seminar for University of Michigan Controls seminar, April 4, 2024 <https://ece.engin.umich.edu/event/invited-speaker-title-and-abstract-forthcoming-21>.

- “Design of wave energy converters using impedance matching principles,” seminar for Virginia Tech Aerospace and Ocean Engineering Department, February 20, 2025 <https://www.aoe.vt.edu/about-us/seminars/Entries/02-20-2025-seminar.html>.
- “Ocean Renewable Energy,” with Jesse Roberts, remote lecture for Duke University, October 21, 2024.
- “WEC Modeling and Controls Workshop,” Oregon State University, Corvallis, OR, May 13–17, 2024, <https://teamer-us.org/workshops/wec-modeling-controls-workshop/>.
- “Using waves to power oceanographic science,” keynote for the North Carolina Renewable Ocean Energy Symposium, Coastal Studies Institute, Wanchese, NC, April 8, 2024.
- “Design, modeling, and control of wave energy converters,” Woods Hole Oceanographic Institute, April 4, 2023.
- “Ocean Renewable Energy,” with Jesse Roberts, remote lecture for Duke University, January 18, 2023.
- “Design, modeling, and control of wave energy converters,” University of Minnesota, October 19, 2022, <https://cse.umn.edu/me/events/departamental-seminar-ryan-coe>.
- “Control Co-design of Wave Energy Converters,” with Giorgio Bacelli, The 16th International Conference on Motion and Vibration Control (MoViC 2022), Sept. 8, 2022.
- “Ocean Renewable Energy,” with Jesse Roberts, remote lecture for Duke University, October 11, 2021.
- “Practical approaches to modeling and control design for a wave energy converter,” with Giorgio Bacelli and Umesh Korde, OMAE2021 short course, June 17-18, 2021, <https://event.asme.org/OMAE-2021/Program/Short-Courses>.
- “Applied WEC modeling and control design,” with Giorgio Bacelli, remote lecture for Virginia Tech. Oct. 29, 2020.
- “Applied WEC modeling and control design,” remote lecture for University of Massachusetts, Amherst, MIE 597CE/ 697CE Ocean Renewable Energy. Oct. 29, 2020.
- “Wave energy converters,” remote lecture for NAVSEA-Carderock summer intern students, June 30, 2020 and July 28, 2020.
- “Integrated WEC design: theory and practice,” with Giorgio Bacelli and Ted Brekken. Pacific Marine Center Workshop. Corvallis, OR. Feb. 3-4, 2020, <https://www.pmec.us/wec-design-workshop>.
- “WPTO Support to Testing,” panel discussion with Jonathan Colby, Steve DeWitt, Lauren Moraski, and Brian Polagye. U.S. Department of Energy’s Water Power Program 2019 Peer Review. Washington, DC. October 8, 2019.
- “Relevance to industry commercialization”, IMREC, Washington, D.C., April 1, 2019.
- “WEC design practices and tools,” with Kelley Ruehl and Yi-Hsiang Yu, OMAE short course. Madrid, Spain. June 16, 2018, <https://archive.asme.org/events/omae2018/program/short-courses>.
- “Environmental characterization and design response analysis for MHK.” DOE Executive Summit on Marine and Hydrokinetic Research and Development. Washington, DC. April 2, 2016.
- “Design response modeling for wave energy converters.” University of Maine. Orono, Maine. April 29, 2016.
- Virginia Tech Future Professoriate Certificate, Virginia Tech, Blacksburg, VA, 2013. <https://graduateschool.vt.edu/transformational-graduate-education-experience/future-professoriate/future-professoriate-certificate.html>
- Teaching assistant, Experiments For Ocean Vehicle Design (AOE4205, AOE4206), Virginia Tech, Blacksburg, VA, 2009-2013.

Mentoring

Post-docs

- Jantzen Lee, 2023-2026
- Alicia Keow, 2021-2026

- Jorge Leon, 2020-2022
R&D engineer, Sandia National Labs
- Daniel Gaebele, 2021-2023
R&D engineer, Sandia National Labs
- Dominic Forbush, 2019-2021
R&D engineer, Sandia National Labs
- Hancheol Cho, 2017-2019
Assistant Professor, Embry-Riddle Aeronautical University

Graduate students

- Becca McCabe (Cornell University, PhD) - committee member, 2023-present
- Jarrah Orphin (University of Tasmania, PhD) – external examiner, 2020
Uncertainty in Hydrodynamic Model Test Experiments of Wave Energy Converters
<https://figshare.utas.edu.au/ndownloader/files/40978802>
- Hannah Mankle (Oregon State University, MS) – committee member, 2019-2021
Advanced Computational Modeling and Design of Wave Energy and Floating Offshore Wind Energy Technologies
https://ir.library.oregonstate.edu/concern/graduate_thesis_or_dissertations/41687q926

Student interns

- Elaine Liu (Stanford University, MS student), 2024
Graduate student, Stanford University
- Andrea Galvan (University of Texas at El Paso, junior), 2023
Graduate student, Industrial and Systems Engineering, Virginia Tech
- Gabriel Fuentes (University of Texas at El Paso, sophomore), 2023
- Anwi Fomukong (University of New Mexico, freshman), 2021-2022
R&D intern, Sandia National Labs
- Sean Pluemer (University of New Mexico, senior), 2020-2021
- Nicholas Ross (University of New Mexico, freshman), 2020-2021
- Delaney Heileman (University of New Mexico, junior), 2019-2021
ORISE fellow
- Sabrina Spracklen (US Naval Academy, junior), 2020
- Preston Rhodes (US Naval Academy, junior), 2020
- John Quinlan (University of New Mexico, junior), 2019
R&D engineer, Sandia National Laboratories
- Francisco Colorbio (University of Texas at El Paso, junior), 2019
Mechanical/test engineer, White Sands Missile Range
- Zach Morrell (University of New Mexico, junior), 2019
Graduate student, University of Chicago
- Bibiana Seng (University of New Mexico, senior), 2017-2018
Quality Assurance Engineer, Fringe
- Tyler Esterly (University of New Mexico, senior), 2016-2017
R&D Software Developer, Sandia National Laboratories
- Jarred Canning (University of Texas at Austin, MS student), 2016-2017
Energy Storage Engineer, RES
- Sam Edwards (University of Michigan, PhD student), 2017
Naval Architect, Naval Surface Warfare Center, Carderock Division (NSWCCD)

- Lori Smith (Virginia Tech, junior), 2017
- Austin Berrier (Virginia Tech, senior), 2017
- Zach Stuart (University of New Mexico, PhD student), 2017
- Chris Sharp (Oregon State University, PhD student), 2016
Associate Professor of Mechanical Engineering at George Fox University
- Alex Campbell (University of Wisconsin-Madison, MS), 2014
Senior Scientist, Areté

Selected research projects

Project	Funding agency	Budget	Year	Description
Pioneer WEC	US Department of Energy	\$7500000	2023	Design and build of a small-scale wave energy converter to provide power to a mooring in the OOI Pioneer Array
WEC-Spar	US Department of Energy	\$500000	2022	Development of a small-scale wave energy converter to provide power to the floating spar buoy
WEC co-design	US Department of Energy	\$10688700	2019	Application of control co-design methods to wave energy converters
WecOptTool	US Department of Energy	\$3276000	2017	Development and application of a wave energy converter design optimization tool utilizing pseudo-spectral methods
Advanced WEC dynamics and controls	US Department of Energy	\$3263499	2016	Design, modeling, and experimental testing of wave energy converters
FOA1310-M3	US Department of Energy	\$130000	2016	Scour modeling and reduction for the M3 wave energy converter
FOA1310-Ecomerit	US Department of Energy	\$20000	2016	Support for Ecomerit wave energy converter modeling and design
FOA1310-Oscilla	US Department of Energy	\$75000	2016	Support for Oscilla wave energy converter modeling and design
Extreme conditions modeling	US Department of Energy	\$1026000	2014	Extreme conditions estimation and design response modeling of wave energy converters

Work experience

- 2026 - present: Associate Professor
Kevin T. Crofton Department of Aerospace and Ocean Engineering, Virginia Tech, Blacksburg, VA
- 2019 - 2026: Principal Member of the Technical Staff
Water Power Technologies Department, Sandia National Labs, Albuquerque, NM
- 2014 - 2019: Senior Member of the Technical Staff
Water Power Technologies Department, Sandia National Labs, Albuquerque, NM
- 2013 - 2014: Postdoctoral Researcher
Water Power Technologies Department, Sandia National Labs, Albuquerque, NM
- 2012: Lead Instructor
Department of Aerospace and Ocean Engineering, Virginia Tech, Blacksburg, VA
- 2009 - 2013: Graduate Research Assistant
Department of Aerospace and Ocean Engineering, Virginia Tech, Blacksburg, VA
- 2009 - 2013: Teaching Assistant
Department of Aerospace and Ocean Engineering, Virginia Tech, Blacksburg, VA
- 2008: Systems Engineer (summer internship)
Northrop Grumman, Newport News, VA
- 2007: Manufacturing Engineer (summer internship)
Northrop Grumman, Newport News, VA

Popular media

- Bench testing a Wave Energy Converter for the Ocean Observatories Initiative Pioneer Array, August 26, 2024 <https://youtu.be/ic-ivEF9eTs?si=ZNPsI-zRR2tTFxU1>
- Co-designing the future of wave energy: Sandia researcher presents at NCROEP, May 22, 2024
<https://energy.sandia.gov/news/sandia-researcher-presents-at-ncroep>
- Sandia makes waves at White House Demo Day March 7, 2024
<https://energy.sandia.gov/news/sandia-makes-waves-at-white-house-demo-day>
- The WaveBot goes to Carderock, February 21, 2024
<https://energy.sandia.gov/news/the-wavebot-goes-to-carderock>
- The Pioneer Array: Harnessing Wave Power to Collect Ocean Data, November 29, 2023
English: <https://youtu.be/PsNv1Gnm59o?si=5yRf28tHVcv-roRM>
Español: https://youtu.be/Tx8KqHqVaDk?si=stCuijs_fLxeenA3
- Sandia Wave Energy Power Take-off (SWEPT) Lab Testing with AquaHarmonics, November 6, 2023
<https://youtu.be/rMaL9X5v-YU?si=QR9400f44INGppwF>
- Wave to wire — a patented approach, September 20, 2023
<https://energy.sandia.gov/news/wave-to-wire-a-patented-approach>
- Sandia Interactive Wave Energy Education Display, September 2, 2021
<https://youtu.be/PcOHZWLtsHc?si=NGp4rf0F3hlPLWnm>
- Celebrating Ocean Month with Ryan Coe, June 22, 2020
<https://www.energy.gov/articles/celebrating-ocean-month-ryan-coe>
- FOSWEC Modeling and Control Design Testing, August 31, 2020
<https://youtu.be/OUxbEC2K6Y>

- Sandia Wave Energy Power Takeoff (SWEPT) Lab, April 22, 2019
<https://youtu.be/IQgPLUfttN8>
- New Lab Offers Mobile, Specialized Testing for Wave Energy Converter Power Take-Off Systems, February 13, 2019
<https://energy.sandia.gov/new-lab-offers-mobile-specialized-testing-for-wave-energy-converter-power-take-off-systems/>
- Advances in WEC Dynamics and Controls, February 6, 2019
https://youtu.be/c4npWk_-Pjk
- Using Robotics to Absorb the Power of Ocean Waves, November 13, 2017
<https://www.ecomagazine.com/news/oceans/using-robotics-to-absorb-the-power-of-ocean-waves>
- Riding along on the crest of a wave, November 3, 2017
<https://portal.engineersaustralia.org.au/news/riding-along-crest-wave>
- Sandia researchers double wave energy converters' efficiency, November 1, 2017
<https://dailyenergyinsider.com/news/8786-sandia-researchers-double-wave-energy-converters-efficiency/>
- Sandia Engineers Increase Efficiency of Wave Energy Converters, October 30, 2017
https://www.renewableenergymagazine.com/ocean_energy/sandia-engineers-increase-efficiency-of-wave-energy-20171030
- Sandia and U.S. Navy Team Up on Wave Energy Testing, August 22, 2017
<https://youtu.be/XCB12Het4c4>
- 360° Video: Wave Energy Testing at the Navys MASK Basin, August 17, 2017
<https://youtu.be/WDuIg8nVtS8>
- Surfing USA in the Name of Science, October 17, 2016
<https://www.afcea.org/content/Blog-surfing-usa-name-science>
- Will a 12 million-gallon tank lead to better wave energy converters?, October 17, 2016
<https://www.ebmag.com/will-a-12-million-gallon-tank-lead-to-better-wave-energy-converters-19216/>
- Wave energy researchers dive deep to advance clean energy source, October 13, 2016
https://www.eurekalert.org/pub_releases/2016-10/ddoe-wer101316.php
- Sandia National Laboratories Uses Its Wave Energy Converter (WEC) to Harness the Motion of the Ocean, August 1, 2016
<https://vimeo.com/177122991>